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Topic 3 — Chemical Changes

Edexcel GCSE (9-1) Chemistry (1CH0) — Paper 1

Topic 3 covers two big families of reactions. The first half is about **acids**: what makes a solution acidic or alkaline, the pH scale, how acids react with metals, bases and carbonates, which salts are soluble, and how to make pure samples of salts. The second half is about **electrolysis**: using electricity to decompose ionic compounds, and predicting what forms at each electrode. Three core practicals sit in this topic: the pH change when calcium oxide or hydroxide is added to hydrochloric acid, preparing copper sulfate crystals, and the electrolysis of copper sulfate solution.

Content marked **(HT)** is Higher Tier only. Topic 3 is shared with GCSE Combined Science; the separate-chemistry-only content of 1CH0 is concentrated in Topics 5 and 9.

3.1 Acids, Alkalis and the pH Scale

Indicators

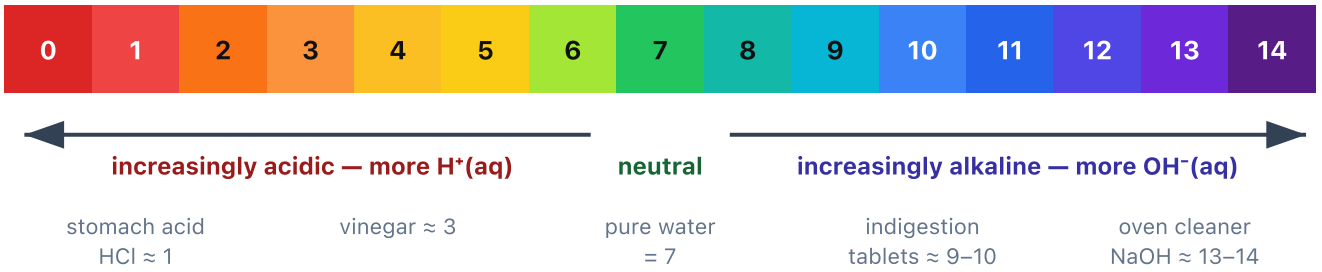
An **indicator** is a dye that changes colour depending on whether it is in an acidic or alkaline solution. You must recall the colours of three single indicators:

Indicator	Colour in acid	Colour in alkali	Notes
Litmus	Red	Blue	Purple in neutral solution; used as paper or solution
Methyl orange	Red	Yellow	Orange at the end point; good for titrations
Phenolphthalein	Colourless	Pink	Sharp colour change; good for titrations

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Universal indicator is a mixture of indicators that shows a range of colours, so it can estimate the **pH** of a solution, not just whether it is acidic or alkaline. A **pH meter** (or pH probe) gives a more accurate numerical reading.

The pH scale and universal indicator colours



pH < 7 acidic; pH = 7 neutral; pH > 7 alkaline.

What makes a solution acidic or alkaline?

Acids are substances that produce **hydrogen ions, $H^+(aq)$**, when dissolved in water.
Alkalis are substances that produce **hydroxide ions, $OH^-(aq)$**, when dissolved in water.

Acid	Formula	Ions in solution	Salts formed
Hydrochloric acid	HCl	$H^+ + Cl^-$	Chlorides
Sulfuric acid	H_2SO_4	$2H^+ + SO_4^{2-}$	Sulfates
Nitric acid	HNO_3	$H^+ + NO_3^-$	Nitrates

- The **higher** the concentration of H^+ ions, the **lower** the pH.
- The **higher** the concentration of OH^- ions, the **higher** the pH.
- **(HT)** pH is a logarithmic scale: each time the **H^+ concentration increases by a factor of 10, the pH decreases by 1**. A solution of pH 2 has 10 times the H^+ concentration of one at pH 3, and 100 times that of one at pH 4.

Worked Example: (HT) Solution A has pH 1 and solution B has pH 4. How many times greater is the H^+ ion concentration in A than in B?

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Difference in pH = 4 – 1 = 3 → factor = 10^3 = **1000 times** greater.

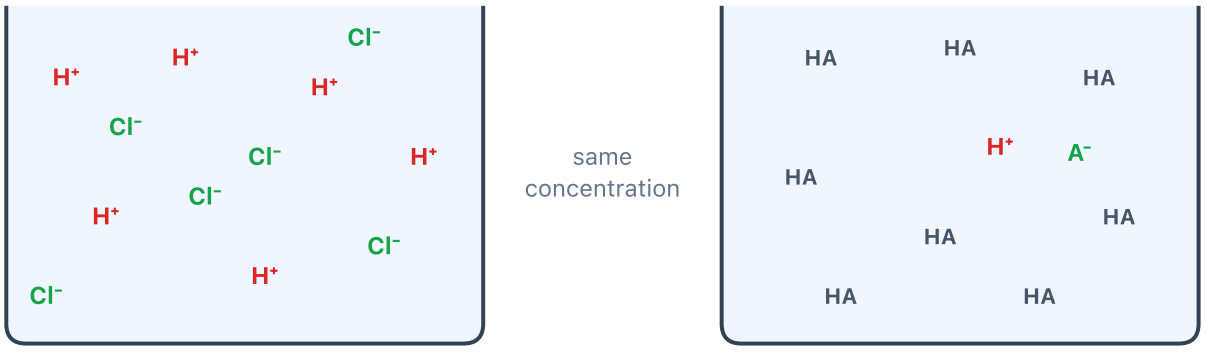
Concentrated and dilute (HT)

Concentration describes the **amount of a substance (solute) in a given volume of solution**.
A **concentrated** acid has a large amount of acid dissolved per unit volume; a **dilute** acid has a small amount of acid per unit volume.

Strong and weak acids (HT)

A **strong acid** is **completely dissociated** (ionised) into ions in aqueous solution — e.g. hydrochloric, nitric and sulfuric acids: $HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$.
A **weak acid** is only **partially dissociated** in aqueous solution — most of its molecules stay un-ionised — e.g. ethanoic, citric and carbonic acids: $CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$.

(HT) Strong vs weak acid at the same concentration



Strong (HCl): fully dissociated | **Weak (e.g. ethanoic): few molecules ionised**

At the **same concentration**, a strong acid has a **higher H^+ concentration** and therefore a **lower pH** than a weak acid. It also reacts faster with metals and carbonates, because there are more H^+ ions to collide with.

	Strong / weak	Concentrated / dilute

Describes	The **proportion** of acid molecules that dissociate into ions	The **amount** of acid dissolved per unit volume
Can be changed by adding water?	No — it is a property of the acid	Yes — adding water dilutes it
Example	Hydrochloric acid is always strong	You can have dilute HCl or concentrated HCl

Exam Tip: Never use "strong" to mean "concentrated". A *dilute* solution of a *strong* acid (e.g. 0.01 mol dm^{-3} HCl) and a *concentrated* solution of a *weak* acid are both possible. In a 4-6 mark answer, define both pairs of terms explicitly in terms of *dissociation* and *amount per volume*.

3.2 Bases, Alkalis and Neutralisation

A **base** is a substance that reacts with an acid to form a salt and water only — typically a **metal oxide** or **metal hydroxide**.
An **alkali** is a base that is **soluble in water** (e.g. sodium hydroxide, potassium hydroxide, calcium hydroxide). All alkalis are bases, but not all bases are alkalis (copper(II) oxide is an insoluble base).

Neutralisation is the reaction between an acid and a base. In terms of ions, the hydrogen ions from the acid react with hydroxide ions from the alkali to make water:

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$$

This ionic equation is the same for *every* acid–alkali neutralisation, whichever acid and alkali are used. The other ions (e.g. Na^+ and Cl^-) are spectator ions that make up the salt.

Reactions of acids — general equations

Reaction	General equation	Example
Acid + metal	acid + metal → salt + hydrogen	$Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$
Acid + metal oxide	acid + metal oxide → salt + water	$CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$
Acid + metal hydroxide	acid + metal hydroxide → salt + water	$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$
Acid + metal carbonate	acid + metal carbonate → salt + water + carbon dioxide	$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

- Only metals **more reactive than hydrogen** react with dilute acids (e.g. Mg, Zn, Fe). Copper, silver and gold do not react. Very reactive metals (sodium, potassium) react dangerously and are never added to acids in school.
- With metals and carbonates you see **effervescence** (fizzing) as a gas is produced, and the solid disappears.
- Naming the salt: first part from the metal (or ammonium), second part from the acid — hydrochloric → **chloride**, sulfuric → **sulfate**, nitric → **nitrate**.

Worked Example: Write a balanced equation for magnesium oxide reacting with nitric acid. Salt: magnesium nitrate, $Mg(NO_3)_2$ (Mg^{2+} and two NO_3^-).
$MgO(s) + 2HNO_3(aq) \rightarrow Mg(NO_3)_2(aq) + H_2O(l)$

Tests for hydrogen and carbon dioxide

Gas	Test	Positive result
Hydrogen, H_2	Hold a **lighted splint** at the mouth of the test tube	Burns with a **squeaky pop** (hydrogen reacts with oxygen to make water)
Carbon dioxide, CO_2	Bubble the gas through **limewater** (calcium hydroxide	Limewater turns **milky / cloudy** (a white precipitate of calcium carbonate forms)

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	solution)	

Limewater reaction: $Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$. Other gas tests you will use alongside electrolysis: **oxygen** relights a **glowing splint**; **chlorine** turns damp blue litmus paper red and then **bleaches** it white.

> **Exam Tip:** Do not mix up the splints: **lighted** splint + squeaky pop = hydrogen; **glowing** splint that relights = oxygen. Carbon dioxide *puts out* a lighted splint, but so do other gases — the only accepted test for CO_2 is limewater.

3.3 Core Practical: pH Change When Calcium Oxide or Calcium Hydroxide is Added to Hydrochloric Acid

Aim: to investigate how the pH of a fixed volume of dilute hydrochloric acid changes as powdered calcium hydroxide (or calcium oxide) is added in small portions.

$$Ca(OH)_2(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + 2H_2O(l)$$

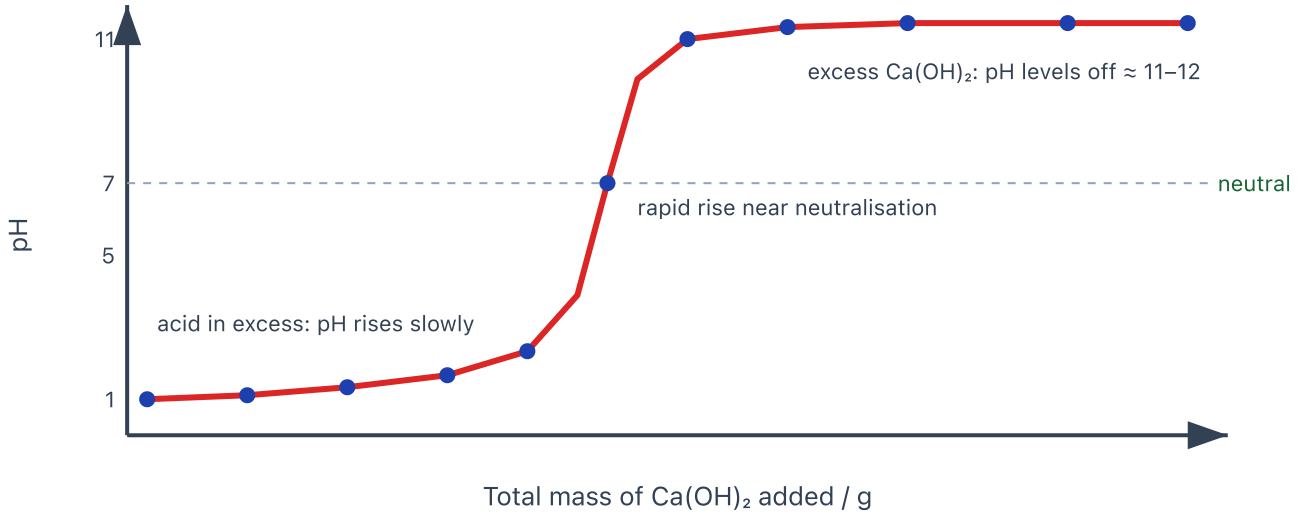
$$CaO(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l)$$

Method

1. Measure **50 cm³** of dilute hydrochloric acid (e.g. 0.1 mol dm^{-3}) using a measuring cylinder and pour into a beaker.
2. Measure the starting pH, using a **pH meter** (calibrated first) or by placing a drop of solution on **universal indicator paper** with a glass rod and comparing with the colour chart.
3. Weigh out about **0.3 g** of calcium hydroxide powder on a mass balance and add it to the acid. **Stir** thoroughly with a glass rod.
4. Record the pH again.
5. Repeat steps 3–4 until about 10 portions have been added and the pH no longer changes.
6. Plot a graph of **pH (y-axis) against total mass of calcium hydroxide added (x-axis)**.

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Typical results: pH against mass of calcium hydroxide added



Analysis and evaluation

- At first the pH rises only slowly, because the acid is in excess and most H^+ ions remain.
- Close to the neutralisation point the pH rises **very steeply** — a small amount of base removes the last H^+ ions.
- After neutralisation, extra calcium hydroxide makes the solution alkaline. The pH levels off at about 12 because calcium hydroxide is only **slightly soluble** — undissolved white solid remains in the beaker.
- The mass needed to neutralise the acid can be read from the steepest part of the graph (where pH = 7).

Issue	Improvement / explanation
Universal indicator paper only gives whole-number pH and colour matching is subjective	Use a calibrated **pH meter** for more precise, objective readings
Powder not fully reacted when pH read	Stir for the same time (e.g. 30 s) after each addition before measuring
Too few points near neutralisation	Use smaller portions (e.g. 0.1 g) near the steep part of the curve

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Hazards	Calcium hydroxide and calcium oxide are irritant/corrosive to eyes — wear eye protection; calcium oxide reacts exothermically with water

3.4 Solubility of Salts and Precipitation

You must recall the general **solubility rules** for common ionic compounds in water:

Compounds	Soluble	Insoluble (main exceptions)
Common **sodium, potassium and ammonium** salts	**All** soluble	—
Nitrates	**All** soluble	—
Common **chlorides**	Soluble	**Silver chloride** and **lead chloride**
Common **sulfates**	Soluble	**Lead, barium and calcium** sulfates
Common **carbonates** and **hydroxides**	Only those of **sodium, potassium and ammonium**	All others insoluble (e.g. $CaCO_3$, $CuCO_3$, $Cu(OH)_2$, $Fe(OH)_3$)

A **precipitate** is an insoluble solid that forms when two solutions are mixed. A **precipitation reaction** happens when the positive ion from one solution and the negative ion from another form an insoluble salt.

Worked Example: Will a precipitate form when barium chloride solution is mixed with sodium sulfate solution? If so, name it.
Possible new combinations: barium sulfate and sodium chloride. Sodium salts are all soluble, but **barium sulfate is insoluble** → a **white precipitate of barium sulfate** forms.

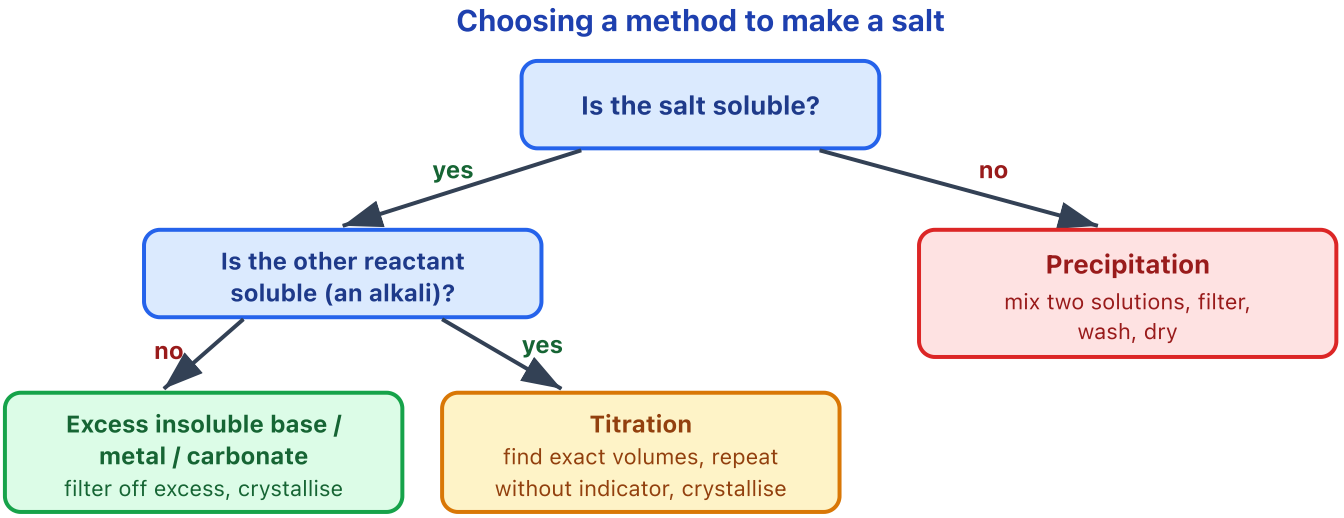
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$BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$ (**(HT)** ionic: $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$)

Exam Tip: To predict a precipitate, swap partners: write down the two "new" combinations of ions and check each against the solubility rules. If either is insoluble, a precipitate forms. Remember that sodium, potassium, ammonium and nitrate compounds are *always* soluble, so they are never the precipitate.

3.5 Making Salts

The method you use to make a salt depends on whether the salt is **soluble** and whether the base you start with is **soluble**.



Method 1: soluble salt from an insoluble reactant (excess solid)

When the base (or metal, or carbonate) is **insoluble**, add it **in excess** to the acid. Using excess makes sure **all the acid reacts**, so the final solution contains only salt and water. The unreacted solid is easy to remove by filtration because it is insoluble.

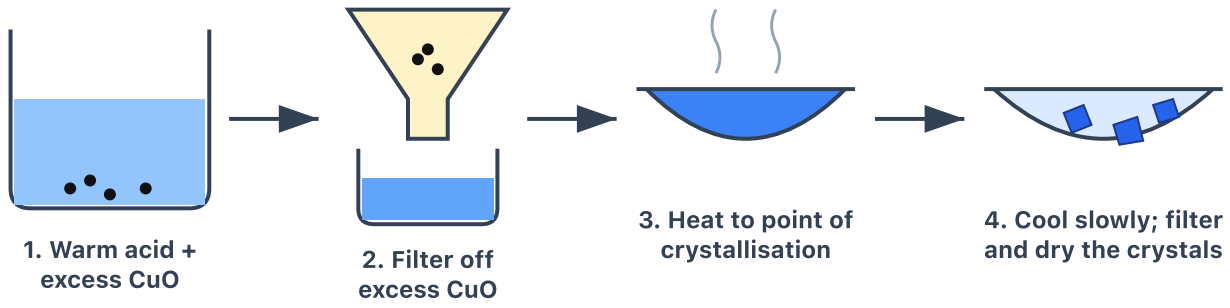
Core Practical: preparing pure, dry hydrated copper sulfate crystals from copper oxide

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$$CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$$

1. Measure about 20 cm^3 of dilute **sulfuric acid** into a beaker and **warm it gently** (e.g. over a water bath or low Bunsen flame) — warming speeds up the reaction. Do not boil.
2. Add **copper(II) oxide** powder, a spatula at a time, stirring with a glass rod. Continue until **no more dissolves** and a little black solid remains — this shows the copper oxide is **in excess** and all the acid has been used up.
3. **Filter** the warm mixture to remove the excess copper oxide. The filtrate is blue copper sulfate solution.
4. Pour the filtrate into an evaporating basin and **heat** it (over a water bath) to evaporate some of the water, until crystals start to form on a cold glass rod dipped in it (point of crystallisation).
5. Leave the basin in a cool place for a day or so so that the solution **crystallises slowly**, forming blue crystals.
6. **Filter** or decant off the remaining liquid, and **dry the crystals** by patting with filter paper (or in a warm oven, not too hot).

Preparing copper sulfate crystals: the four stages



Step	Reason
Warm the acid	Increases the rate of reaction
Add copper oxide in excess	Ensures all the acid reacts — no acid left in the product
Filter	Removes unreacted (insoluble) copper oxide

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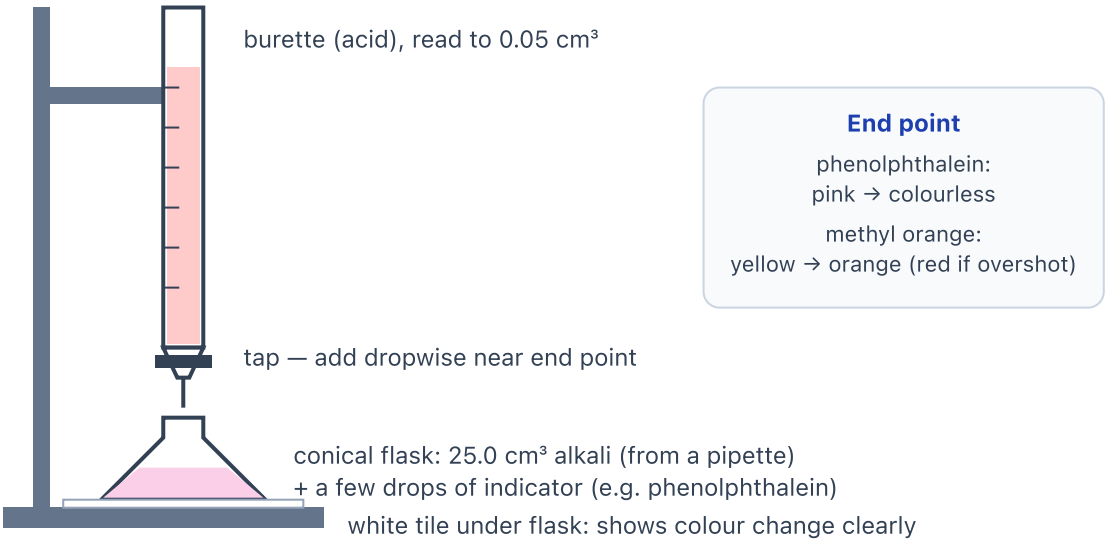
Heat only until crystallisation point; do not evaporate to dryness	Gives larger, purer crystals; keeps the water of crystallisation ($CuSO_4 \cdot 5H_2O$ is blue; overheating gives white anhydrous $CuSO_4$)
Dry with filter paper / low-temperature oven	Removes surface water without decomposing the hydrated crystals

Exam Tip: Hazards: sulfuric acid is an irritant (corrosive if concentrated) and copper oxide and copper sulfate are harmful — wear eye protection, and avoid spitting by heating gently. If asked why the acid is warmed rather than boiled: boiling would cause the acid to spit and could make it more concentrated.

Method 2: soluble salt from an acid and an alkali — titration

When both reactants are **soluble**, you cannot add "excess" and filter it off, because the excess would stay dissolved and contaminate the salt. Instead you must add the **exact** volume of acid needed to neutralise the alkali, found by **titration**. This method is used to make sodium, potassium and ammonium salts.

Titration apparatus



1. Use a **pipette** (with pipette filler) to measure exactly 25.0 cm³ of alkali (e.g. sodium hydroxide) into a conical flask. Add a few drops of a suitable indicator (phenolphthalein or methyl orange — *not* universal indicator, whose gradual colour change gives no sharp end point).

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2. Fill a **burette** with the acid and record the starting volume (read the bottom of the meniscus at eye level).
3. Add the acid to the alkali while **swirling** the flask, adding **drop by drop** near the end point, until the indicator just changes colour. Record the final burette reading; volume added = final – initial (the **titre**).
4. Repeat to get **concordant** results (titres within 0.10 cm^3 of each other) and calculate the mean.
5. Now **repeat the titration without the indicator**, adding exactly the mean volume of acid to a fresh 25.0 cm^3 of alkali. (Alternatively, remove the indicator by adding activated charcoal and filtering.)
6. Heat the resulting salt solution to the point of crystallisation, leave it to cool and crystallise, then filter and dry the crystals.

> **Exam Tip:** Why must the titration be repeated without indicator? The indicator is a dye that would otherwise *contaminate* the salt crystals. Why can't you add excess alkali and filter? Because the alkali is *soluble* and cannot be removed by filtering — the salt would be impure.

Method 3: insoluble salts — precipitation

An **insoluble salt** is made by mixing two solutions, one containing its positive ion and one containing its negative ion. For example, to make **silver chloride** (or lead iodide, barium sulfate, calcium carbonate):

$$AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$$

1. Dissolve each soluble reactant in **distilled (deionised) water** — tap water contains ions that could form other precipitates.
2. **Mix** the two solutions in a beaker and stir. A precipitate forms immediately.
3. **Filter** the mixture; the insoluble salt is the residue on the filter paper.
4. **Wash** the residue on the filter paper with distilled water to remove the soluble impurities (the other salt, e.g. $NaNO_3$, and any excess reactant).
5. **Dry** the solid in a warm oven or by leaving it on filter paper.

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Worked Example: Choose two solutions to make barium sulfate by precipitation.
Need a soluble barium compound and a soluble sulfate: e.g. **barium chloride** (or barium nitrate) and **sodium sulfate** (or dilute sulfuric acid).
$BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$. Filter, wash with distilled water, dry.

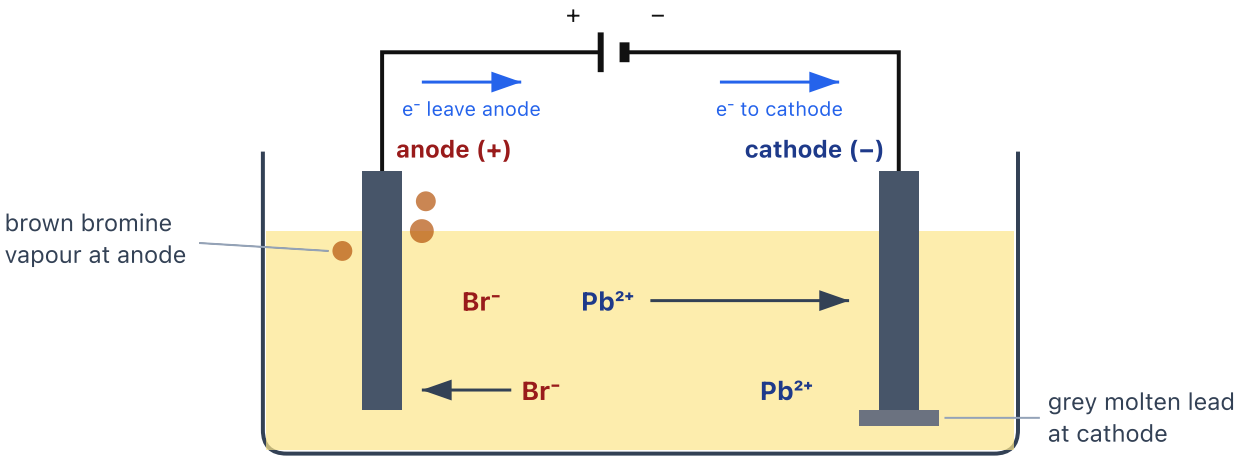
3.6 Electrolysis

An **electrolyte** is an **ionic compound** that is **molten** or **dissolved in water**, so that its ions are free to move and it can conduct electricity.
Electrolysis is the decomposition of an electrolyte using a direct current of electricity.

- The negative electrode is the **cathode**; the positive electrode is the **anode**.
- Positive ions (**cations**) are attracted to the cathode. Negative ions (**anions**) are attracted to the anode.
- When ions reach the electrodes they are **discharged**: cations **gain** electrons at the cathode and anions **lose** electrons at the anode, forming elements.
- **Inert electrodes** (graphite or platinum) do not react; they just conduct electricity into and out of the electrolyte.
- Solid ionic compounds cannot be electrolysed — the ions are fixed in the lattice and cannot move.

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Electrolysis of molten lead bromide (inert electrodes)



Heat is needed to melt the lead bromide so its ions can move. Done in a fume cupboard (bromine is toxic).

Electrolysis of molten compounds

A molten binary ionic compound contains only two kinds of ion, so the products are simple: the **metal** forms at the cathode and the **non-metal** forms at the anode.

lead bromide → lead + bromine $PbBr_2(l) \rightarrow Pb(l) + Br_2(g)$

Electrolysis of aqueous solutions

In an aqueous solution, water also provides ions: a very small proportion of water molecules ionise, $H_2O \rightleftharpoons H^+ + OH^-$. So there are **two cations** (the metal ion and H^+) competing at the cathode and **two anions** (the non-metal ion and OH^-) competing at the anode. Only one of each is discharged:

Electrode	Rule
Cathode (−)	**Hydrogen** is produced, *unless* the metal is **less reactive than hydrogen** (e.g. copper, silver), in which case the **metal** is deposited.
Anode (+)	**Oxygen** is produced, *unless* the solution contains **halide ions** (Cl^-, Br^-, I^-), in which case the **halogen** is produced. (Sulfate and nitrate ions are never discharged.)

Electrolytes you must know (inert electrodes)

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Electrolyte	Ions present	Cathode product	Anode product	Left in solution / observations
Molten lead bromide, $PbBr_2(l)$	Pb^{2+}, Br^-	Lead	Bromine	Grey liquid metal; brown vapour
Copper chloride solution, $CuCl_2(aq)$	Cu^{2+}, H^+, Cl^-, OH^-	**Copper** (less reactive than H)	**Chlorine** (halide present)	Pink-brown coating on cathode; bubbles that bleach damp litmus; blue colour fades
Sodium chloride solution, NaCl(aq)	Na^+, H^+, Cl^-, OH^-	**Hydrogen** (Na more reactive than H)	**Chlorine**	Sodium hydroxide solution is left
Sodium sulfate solution, $Na_2SO_4(aq)$	Na^+, H^+, SO_4^{2-}, OH^-	**Hydrogen**	**Oxygen**	Only water is decomposed; sodium sulfate left unchanged
Water acidified with sulfuric acid	H^+, SO_4^{2-}, OH^-	**Hydrogen**	**Oxygen**	Volume of H_2 is **twice** the volume of O_2 ($2H_2O \rightarrow 2H_2 + O_2$)
Copper sulfate solution, $CuSO_4(aq)$	Cu^{2+}, H^+, SO_4^{2-}, OH^-	**Copper**	**Oxygen**	Blue colour fades as Cu^{2+} is removed; solution becomes acidic (H^+ and SO_4^{2-} left = sulfuric acid)

Exam Tip: Two quick checks: pure water conducts very poorly, so a little sulfuric acid (or sodium sulfate) is added to provide ions — the acid itself is not used up. And in NaCl(aq), sodium is *never* formed at the cathode in solution; hydrogen is, because sodium is more reactive than hydrogen.

Oxidation, reduction and half equations (HT)

Oxidation is loss of electrons; reduction is gain of electrons (OIL RIG).
In electrolysis, **reduction** always happens at the **cathode** (positive ions gain electrons) and **oxidation** always happens at the **anode** (negative ions lose electrons).

A **half equation** shows what happens at one electrode, including electrons (e^-). Both atoms and charges must balance.

Electrode	Half equation	Process
Cathode — lead	$Pb^{2+} + 2e^- \rightarrow Pb$	Reduction
Cathode — copper	$Cu^{2+} + 2e^- \rightarrow Cu$	Reduction
Cathode — hydrogen	$2H^+ + 2e^- \rightarrow H_2$	Reduction
Anode — bromine	$2Br^- \rightarrow Br_2 + 2e^-$	Oxidation
Anode — chlorine	$2Cl^- \rightarrow Cl_2 + 2e^-$	Oxidation
Anode — oxygen	$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$	Oxidation
Anode — copper electrode dissolving	$Cu \rightarrow Cu^{2+} + 2e^-$	Oxidation

Worked Example: (HT) Write the half equation for the formation of aluminium at the cathode during electrolysis of molten aluminium oxide.
Aluminium ions are Al^{3+}, so each needs 3 electrons: **$Al^{3+} + 3e^- \rightarrow Al$** (reduction — electrons gained).

Exam Tip: Electrons appear on the *left* of a cathode half equation (gained) and on the *right* of an anode half equation (lost). Check charge: in $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$ the left is 4– and the right is 4– ✓.

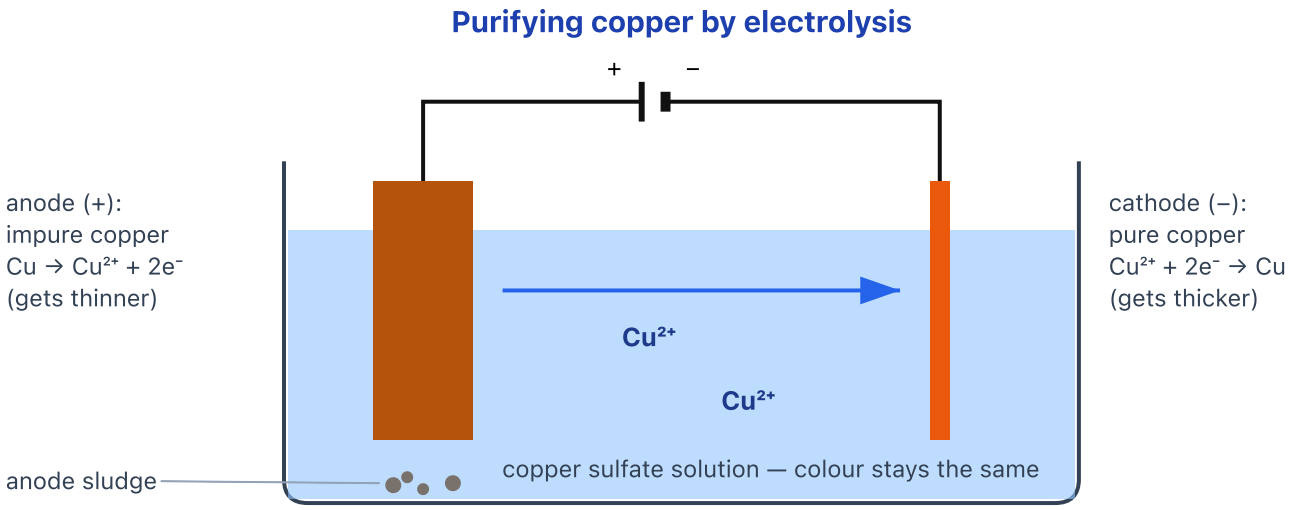
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Electrolysis with copper electrodes — purifying copper

If copper sulfate solution is electrolysed using **copper electrodes** (not inert ones), the anode itself takes part:

- At the **anode**, copper atoms lose electrons and go into solution as Cu^{2+} ions (**(HT)** $Cu \rightarrow Cu^{2+} + 2e^-$). The anode gets **lighter**.
- At the **cathode**, Cu^{2+} ions gain electrons and copper is deposited (**(HT)** $Cu^{2+} + 2e^- \rightarrow Cu$). The cathode gets **heavier**.
- Copper ions are added at the anode at the same rate they are removed at the cathode, so the **concentration and blue colour of the solution stay the same**.
- Ideally, the **mass lost by the anode equals the mass gained by the cathode**.

This is how copper is **purified** industrially: the anode is a block of **impure copper** and the cathode is a thin sheet of **pure copper**. Copper from the anode transfers to the cathode; impurities do not dissolve and fall to the bottom as **anode sludge** (which can contain valuable metals such as silver and gold). In this case the anode loses more mass than the cathode gains.



3.7 Core Practical: Electrolysis of Copper Sulfate Solution

Aim: to investigate the electrolysis of copper sulfate solution using (a) inert (graphite) electrodes and (b) copper electrodes, and how the mass of copper electrodes changes with current.

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Part A — inert (graphite) electrodes

1. Pour copper sulfate solution into a beaker. Place two graphite rods into it, not touching, held in a lid or electrode holder.
2. Connect to a low-voltage d.c. supply (e.g. 4 V) and switch on.
3. Observe: a **pink-brown deposit of copper** forms on the cathode; **bubbles of oxygen** form at the anode (test with a glowing splint, which relights).

Part B — copper electrodes

1. Clean two copper strips with emery paper, wash with propanone (flammable — no flames) or water, dry, and **weigh** each one. Label anode and cathode.
2. Set up the circuit with an **ammeter** and a **variable resistor** in series with the cell. Adjust the resistor so the current is a set value (e.g. 0.2 A).
3. Electrolyse for a fixed time (e.g. 20 minutes), keeping the current constant.
4. Remove the electrodes, rinse **gently** with distilled water then propanone, allow to dry, and **reweigh**.
5. Repeat with fresh electrodes at different currents (e.g. 0.3, 0.4, 0.5 A).
6. Calculate the change in mass of each electrode; plot **change in mass against current**.

Variable	In this practical
Independent	Current (A)
Dependent	Change in mass of each electrode (g)
Control	Time, concentration and volume of solution, size of electrodes and distance between them, temperature

Expected results

- The **cathode gains mass** and the **anode loses mass**; the two changes are about equal.
- The change in mass is **directly proportional to the current** (a straight line through the origin), because the amount of copper transferred is proportional to the charge passed (charge = current × time).

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Worked Example: In 20 minutes at 0.40 A, a copper cathode increases from 12.45 g to 12.61 g, and the anode decreases from 12.50 g to 12.33 g. Calculate the changes and comment.
Cathode: 12.61 – 12.45 = **+0.16 g**; anode: 12.33 – 12.50 = **–0.17 g**.
The anode loss is slightly larger than the cathode gain — some copper may have flaked off the anode or been lost from the cathode while rinsing/drying.

Exam Tip: Common sources of error to evaluate: copper deposit flaking off the cathode when rinsing (rinse gently, don't wipe); electrodes not completely dry when reweighed (dry thoroughly — water adds mass); current drifting during the run (keep adjusting the variable resistor). Hazards: copper sulfate is harmful and an irritant; propanone is highly flammable.

Key Points Summary

- Indicators: litmus (red/blue), methyl orange (red/yellow), phenolphthalein (colourless/pink). Universal indicator or a pH meter measures pH.
- Acids produce $H^+(aq)$; alkalis produce $OH^-(aq)$. More H^+ → lower pH; more OH^- → higher pH.
- **(HT)** Factor of 10 change in $[H^+]$ = 1 pH unit. Concentrated/dilute = amount per volume; strong/weak = complete/partial dissociation.
- Bases neutralise acids to form a salt and water; alkalis are soluble bases. Neutralisation: $H^+ + OH^- \rightarrow H_2O$.
- Acid + metal → salt + hydrogen; acid + metal oxide/hydroxide → salt + water; acid + carbonate → salt + water + CO_2.
- Tests: hydrogen — lighted splint, squeaky pop; carbon dioxide — limewater turns milky.
- Core practical: adding $Ca(OH)_2$/CaO to HCl — pH rises slowly, then steeply, then levels off at about 12.
- Solubility: Na, K, NH_4^+ salts and nitrates soluble; chlorides soluble except Ag and Pb; sulfates soluble except Pb, Ba, Ca; carbonates and hydroxides insoluble except Na, K, NH_4^+.
- Soluble salt from insoluble base: excess solid, filter, crystallise (core practical: $CuSO_4 \cdot 5H_2O$ from CuO). From an alkali: titration, repeat without indicator, crystallise. Insoluble salt: precipitation, filter, wash, dry.

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- Electrolysis: electrolyte = molten or dissolved ionic compound; cations to cathode (–), anions to anode (+).
- Aqueous solutions: cathode gives H_2 unless the metal is less reactive than hydrogen; anode gives O_2 unless halide ions are present.
- Know the products for $PbBr_2(l)$, $CuCl_2(aq)$, $NaCl(aq)$, $Na_2SO_4(aq)$, acidified water and $CuSO_4(aq)$.
- **(HT)** Reduction (gain of e^-) at the cathode; oxidation (loss of e^-) at the anode; write balanced half equations.
- Copper electrodes: anode dissolves, cathode gains copper, solution colour unchanged — used to purify copper. Core practical: mass change $\propto$ current.

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